

INFORMATIC PROJECT

Student 2nd year

Analysis report : NEO-Watch

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1. Introduction and Context

2. Requirements Analysis and Main Features

2.1. Functional Requirements (User Management, Dashboard, Alerts, NASA API).

2.2. Application Features (Use Case Diagram).

```
---
config:
  theme: redux
---
flowchart LR
    Visitor["Visitor"]
    User["User"]
    Admin["Administrator"]

    subgraph subgroup["NEO-watch application"]
        logout([Logout])
        search([Search NEOs])
        download([Download the results])
        favlist([View the list of favorites NEOs])
        addfav([Add a favorite NEO])
        editlist([Edit the list])
        alerts([View alerts])
        editalerts([Edit alerts])
        add([Add NEO])
        update([Update the data])
        manage([Manage accounts])
        accounts([View accounts])
        deleteaccount([Delete an account])
        login([Login])
        signup([Signup])
    end

    end

    User --> search
    User --> favlist
    User --> add
    User --> alerts
    User --> logout
    Admin --> manage
    Admin --> update
    Admin --> User
```

```

Visitor --> signup
Visitor --> login

alerts -. "include" .-> editalerts
manage -. "include" .-> accounts
search -. "extend" .-> download
search -. "extend" .-> addfav
favlist -. "extend" .-> editlist
manage -. "extend" .-> deleteaccount

```

2.3. Application Workflow (Activity Diagram + Settings Activity Diagram).

```

stateDiagram
    main_app:Main application
    search_neo:Search for a NEO
    search_characteristics:Search by characteristics
    download:Download the results
    add_favorites:Add a new NEO as a favorite
    favs_list:View the list of favorites NEOs
    edit_list>Edit the list
    add_neo:Add NEO
    enter_characteristics:Enter characteristics
    New_neo:New NEO
    Delete_user_info>Delete user information
    create_alerts>Create alerts
    settings:Settings
    login:Login
    signup:Signup

    state if_state <<choice>>
    state if_statemain <<choice>>

    [*] --> Home
    Home --> quit
    quit --> [*]
    Home --> signup
    Home --> login
    login --> if_state
    if_state --> main_app:correct password
    if_state --> login:false password

    state main_app {
        [*] --> search_neo
        search_neo --> search_characteristics
        search_characteristics --> add_favorites
        add_favorites --> download
        search_characteristics --> download
        [*] --> favs_list
        favs_list --> edit_list
        [*] --> create_alerts
        [*] --> add_neo
        add_neo --> enter_characteristics
        enter_characteristics --> if_statemain
        if_statemain --> New_neo:New NEO validate
    }

```

```

    if_statemain --> Delete_user_info:NEO already exist
    [*] --> settings
}

```

stateDiagram

```

settings:Settings
logout:Logout
settings_admin:Settings for admin
manage_accounts:Manage accounts
view_accounts:View accounts
delete_accounts>Delete an account
update_data:Update the data
personnal_settings: Personnal settings
change_theme_colors: Change theme color
change_dashbord_organization: Personalized dashboard

state if_state <<choice>>
state if_statemain <<choice>>

state main_app {
    settings --> personnal_settings
    personnal_settings --> change_theme_colors
    personnal_settings --> change_dashbord_organization

    settings --> logout
    settings --> update_data
    settings --> manage_accounts
    logout --> [*]:back to Home
    state settings_admin {
        manage_accounts --> view_accounts
        manage_accounts --> delete_accounts
        update_data
    }
}
}

```

3. Technical Design and Data Structure

In introduction put here the technical characteristics

3.1. Class Architecture (Class Diagram).

3.2. Table Organization (Database Diagram).

```
classDiagram
    USERS "1" --> "0..n" ALERT
    USERS "1" --> "1" LOGIN_HISTORY
    USERS "1" --> "0..n" NEO
    USERS "1" --> "0..n" FAVORITE
    USERS "1" --> "0..n" NOTIFICATION
    NEO "1" --> "0..n" CLOSE_APPROACH
    NEO "0..n" --> "0..n" FAVORITE
    NEO "0..n" --> "0..n" ALERT
    FAVORITE "1" --> "1" DISTANCE_HISTORY
    ALERT "1" --> "0..n" NOTIFICATION

class USERS {
    int id PK
    string email
    string password_hash
    string role
    datetime created_at
}
class LOGIN_HISTORY {
    int id PK
    int user_id FK
    datetime login_at
    string ip_address
}
class NEO {
    int id PK
    string nasa_id
    string name
    float diameter_min_m
    float diameter_max_m
    float absolute_magnitude
    boolean is_hazardous
    boolean is_custom
    int created_by_user_id FK
}
class CLOSE_APPROACH {
    int id PK
```

```

        int neo_id FK
        date approach_date
        float miss_distance_km
        float relative_velocity_kmh
    }
    class FAVORITE {
        int id PK
        int user_id FK
        int neo_id FK
        datetime added_at
    }
    class DISTANCE_HISTORY {
        int id PK
        int favorite_id FK
        date recorded_at
        float distance_km
    }
    class ALERT {
        int id PK
        int user_id FK
        int neo_id FK
        float min_size_m
        float max_distance_km
        boolean is_active
        datetime created_at
    }
    class NOTIFICATION {
        int id PK
        int user_id FK
        int alert_id FK
        string message
        datetime sent_at
    }
}

```


4. Modeling the Application's Layered Structure

4.1. Authentication and Role Management (Login Sequence Diagram)

```
sequenceDiagram
    actor U as User
    participant F as Frontend
    participant B as Backend (FastAPI)
    participant DB as PostgreSQL
    U->>F: Enters login + password
    F->>B: POST /login (credentials)
    B->>DB: Search for user
    DB-->>B: User + hashed password

    alt Valid credentials
        B->>B: Verify password
        B-->>F: JWT token + success
        F-->>U: Display dashboard
    else Invalid credentials
        B-->>F: Error (401 - Unauthorized)
        F-->>U: Display "incorrect credentials"
    end
end
```

4.2. NEO Search Management (Search NEO Sequence Diagram)

```
sequenceDiagram
    actor U as User
    participant F as Frontend
    participant B as Backend (FastAPI)
    participant DB as PostgreSQL

    U->>F: Enters search criteria
    F->>B: GET /neos (criteria + sorting)
    B->>DB: Search matching NEOs
    DB-->>B: List of NEOs

    alt Results exist
        B-->>F: List of NEOs
        F-->>U: Display sorted results
    else No results
        B-->>F: Empty list
        F-->>U: Display "no NEO found"
    end
end
```

4.3. Synchronization with the NASA API (Database Update Sequence Diagram)

```
sequenceDiagram
    actor A as Administrator
    participant F as Frontend
    participant B as Backend (FastAPI)
    participant DB as PostgreSQL
    participant N as NASA API

    A->>F: Clicks "Update data"
    F->>B: POST /admin/refresh (date range)
    B->>N: GET feed (date range)

    alt NASA responds
        N-->>B: List of NEOs
        B->>DB: Update only NASA-origin NEOs
        DB-->>B: Confirmation
        B-->>F: Update completed
        F-->>A: Display update result
    else NASA does not respond
        B-->>F: Update failed
        F-->>A: Display "NASA service unavailable"
    end
end
```

5. User Interface (Menu Diagram)

```
---
config:
    htmlLabels : false
---
flowchart TD
    home_menu["`**Home menu**  
Sign up  
Log in  
Exit`"]
    main_app["`**Main app menu**  
Search for a NEO  
View the dashboard  
Add a new NEO  
View alerts  
View settings`"]
    search_menu["`**Search menu**  
Filter by id  
Filter by name  
Filter by absolute magnitude  
Filter by estimated diameters  
Filter by danger level  
Filter by close approach date  
Filter by close approach distance  
Return to main menu`"]
    settings_user["`**Settings menu** (user)  
Personalized dashboard  
Change theme color  
Logout  
Return to main menu`"]
    settings_admin["`**Settings menu** (admin)  
Update the data"]

    home_menu --> main_app
    main_app --> search_menu
    search_menu --> main_app
    main_app --> settings_user
    settings_user --> main_app
    settings_admin --> settings_user
```

```
View accounts
Delete an account
Return to main menu`]

main_app --> settings_admin
settings_admin --> main_app
```

6. Planning and Project Management (Gantt Chart)

```
---
config:
  theme: redux
---
gantt
  title NEO-Watch Project
  dateFormat YYYY-MM-DD
  section Analysis
    Mentoring 1           :milestone, 2026-08-28, 1d
    Github config         :2026-08-28, 16d
    Introduction phase    :2026-08-28, 8d
    Class diagram         :2026-08-28, 8d
    Mentoring 2           :milestone, 2026-09-04, 1d
    Sequency diagram      :2026-09-04, 8d
    Activity diagram      :2026-09-04, 8d
    Data table            :2026-09-04, 8d
    Gant diagram          :2026-09-04, 8d
    Mentoring 3           :milestone, 2026-09-11, 1d
    Writing               :2026-09-11, 7d
    proofreading          :2026-09-15, 3d
    Analysis document     :milestone, 2026-09-17, 1d
  section Implementation
    class implementation: 2026-09-17, 28d
    Unit Test            :2026-09-17, 28d
    Mentoring 4          :milestone, 2026-09-25, 1d
    DAO implementation   :2026-10-07, 14d
    Data importation     :2026-10-07, 30d
    Optimisation         :2026-10-22, 15d
    Immersion session    :2026-11-03, 3d
  section Final report
    Writing final report :2026-11-03, 15d
    Report proofreading  :2026-11-18, 4d
    Mentoring 7         :milestone, 2026-11-20, 1d
    Submit report       :milestone, 2026-11-21, 1d
    Wrinting presentation :2026-11-22, 12d
    Training presentation :2026-12-05, 5d
    Final presentation  :milestone, 2026-12-09, 1d
```

7. Conclusion and Perspectives